

Demonstration of Proposed Features

Date	12 July 2026
Team ID	[Your Team ID]
Project Name	HDI Predict – Human Development Index Prediction & Insights Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase for the HDI Predict project and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Very High HDI Prediction (Developed Nations)	Accepts life expectancy, mean years of schooling, expected years of schooling and GNI per capita as inputs and predicts a Very High HDI classification for countries with strong performance across all indicators, using a trained ML model.	Implemented	Yes	Validated using sample data for high-development countries; classification accuracy verified.
2	Development Gap Identification (Emerging Economies)	Accepts mid-range indicator values representing a developing nation and predicts a Medium HDI score, highlighting specific dimensions (health, education, income) where improvement is needed.	Implemented	Yes	Tested with moderate-value datasets; gap insights displayed alongside predicted score.
3	Low HDI & Development Intervention Assessment	Evaluates countries with low life expectancy, limited educational attainment and low GNI per capita, predicts a Low HDI score, and flags key developmental challenges for policy intervention.	Implemented	Yes	Predicted output cross-checked against known low-HDI country benchmarks.
4	Data Input & Validation Module	Provides a form-based interface for entering life expectancy, schooling and GNI per capita values, with range checks to ensure statistically valid inputs before prediction.	Implemented	Yes	Input boundaries cross-checked against historical HDI dataset ranges.
5	Four-Tier Classification Engine	Classifies the predicted HDI score into one of four development tiers — Very High, High, Medium or Low — based on standard UNDP threshold bands.	Implemented	Yes	Threshold bands verified against official UNDP

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					classification cut-offs.
6	Research & Policy Analytics Dashboard	Visual dashboard for researchers and policymakers to compare HDI trends, indicator contributions and regional patterns across countries for policy planning.	Partial	No	Core charts implemented; cross-country comparison filters planned for next iteration.
7	Result Visualization & Report Export	Displays predicted HDI score, tier classification and indicator breakdown with visual indicators, and allows users to export a summary report of the analysis.	Partial	No	Visualization logic implemented; export-to-PDF integration pending for live demo.

Feature Implementation Summary:

Total Features Proposed	7
Total Features Implemented	5
Total Features Demonstrated	5
Overall Implementation Rate (%)	71%